

CSCE 581: Introduction to Trusted AI

Lectures 21 and 22: Supervised ML (Text Processing), Trust Issues

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

31ST MARCH AND 2ND APRIL, 2026

**Carolinian Creed: “I will practice personal and
academic integrity.”**

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Organization of Lectures 21, 22

- Introduction Section
 - Recap from Week 10 (Lectures 19 and 20)
 - Announcements and News
- Main Section
 - L21: Supervised ML (Text)
 - L22: Classification, Trust Issue
- Concluding Section
 - About next week – Lectures 23, 24
 - Ask me anything

Recap from Week 10 (Lectures 19, 20)

- We looked at
 - L19 AI - Unstructured (Text): Representation, Common NLP Tasks
 - L20: Natural Languages/ Language Models and their Impact on Text/ AI

AI Trust: News

- Browser extensions of LLM-based chatbots leak information. “In our study, we emphasize that not only do these assistants collect users' highly sensitive data but that this information is also shared with their own servers as well as third-party trackers, which can be further utilized to target highly personalized and sensitive ads to the user.”

Table 1: Overview of studied AI browser assistants sorted by their popularity.

Legend: **Personal Data** : Personally Identifiable Information (**PII**), Personal Communications (**PC**), Financial Information (**FI**).

Web Data : User Activity (**UA**), Web History (**WH**), Website Content (**WC**), Location (**LOC**). **No Data** : No Data collected.

Extension Name	Install Counts	Supported Model(s)	Default Model	Invocation Mode	Response Mode	Data Disclosures	SDK Version
Sider : ChatGPT Sidebar	4M	   	sider	Automatic	Server-side	PII WC	4.35.0
Monica - Your AI Copilot	2M	   	gpt-4o-mini	Mixed	Server-side	PII UA PC FI	7.6.0
ChatGPT for Google	2M	   	gpt-4o-mini	Mixed	Client-side	PII UA PC FI	5.5.1
Merlin Ask AI	1M	   	gpt-4o	Mixed	Server-side	PII LOC	7.3.2
MaxAI : Chat with Webpage	800K	   	gpt-4o-mini	Manual	Server-side	PII UA	6.7.1
Perplexity - AI Companion	500K	  	perplexity	Manual	Server-side	No Data	1.0.21
HARPA AI	400K	  	harpa-v1-smart	Manual	Server-side	PII UA WH WC	9.6.2
Wiseone - AI Copilot	90K	 	gpt-4o	Manual	Server-side	PII WC	1.7.2
TinaMind - AI Assistant	50K	  	gemini-1.5-pro	Manual	Server-side	PII UA PC	2.14.2
Copilot : AI Assistant	30K	  	gpt-4o-mini	Automatic	Server-side	PII	1.5.73

- Article - https://www.theregister.com/2026/03/25/generative_ai_browser_extensions_privacy/
- Paper - Big Help or Big Brother? Auditing Tracking, Profiling, and Personalization in Generative AI Assistants, <https://arxiv.org/abs/2503.16586>, 2026

AI Trust: News

Table 2: Data collection and exfiltration behavior of assistants in public and private online spaces of a user. *Exfiltration legend:*

Full Webpage : Page text, title, location, hyperlinks. **Server-fetch Webpage** : Page title, location, server-fetched file's upload location. **Plain Webpage** : Page text, title, location. **Partial Webpage** : Partial content or missing details. *Response legend:* ✓: Response with Relevant Details. ✗: Missing some details in Response. ⚡: Response restricted. ✖: No response generated.

	Category	WebPage	Sider	Monica	CFG	Merlin	MaxAI	Perplexity	Harpa	Wiscone	TinaMind	Copilot
Public Spaces	News Platforms	cnn.com	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗
	Open Forums	reddit.com	✗	✓	✗	✓	✓	✗	✓	✗	✗	✓
	Informative Articles	wikipedia.org	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗
	E-commerce Website	amazon.com	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗
	Sports Websites	espn.com	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
	Travel Platforms	expedia.com	✓	✓	✓	✓	✓	✗	✓	✗	✓	✓
	User-generated Media	youtube.com	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
	Kids Website	nickjr.com	✗	✓	✓	✓	✓	✓	✓	✗	✓	✓
	Misinformation Website	infowars.com	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Private Spaces	Violence Material	guns.com	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗
	Health Portal	university health portal	✓	✓	✓	✓	✓	✗	✗	✓	✗	✗
	Email Account	mail.google.com	✓	✓	✗	✓	✓	✗	✗	✓	✓	✗
	Social Media Platform	facebook.com	✓	✓	✓	✓	✓	✗	✓	✓	✗	✗
	Adult Content	pornhub.com	✗	✓	✓	✓	✓	⚡	✗	⚡	✓	✗
	Online Streaming Service	netflix.com	✗	✓	✓	✓	✓	✗	✗	✓	✓	✗
	Government Website	irs.gov	✗	✓	✓	✓	✓	✗	✓	✗	✓	✗
	Dating Service	tinder.com	✗	✓	✓	✓	✓	✗	✓	✗	✓	✗
	Financial Service	chase.com	✗	✓	✓	✓	✓	✗	✓	✗	✗	✗
	Educational Platform	canvas.instructure.com	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓
	Messaging Platform	slack.com	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗

“One of the most shocking findings was that GenAI browser assistants were freely able to collect and share data to their own servers on authenticated **health** portals. They were able to answer follow-up questions ranging from patient details to entire medical history. Collection of PHI without appropriate user consent is in clear violation of HIPAA [14]. “

“Moreover, **student's academic records** including assessment scores, exam performances, overall grades – were all collected and shared with browser assistant's servers demonstrating violation of FERPA [34] that aims to protect these attributes for a student.”

AI Trust News: LLM-based Summaries

Bloomberg Has a Rocky Start With A.I. Summaries

- https://www.nytimes.com/2026/03/29/business/media/bloomberg-ai-summaries.html?unlocked_article_code=1.7k4.oSbJ.htksCZb-DQ3E&smid=nytcore-android-share
- The newspaper chain Gannett [uses](#) similar A.I.-generated summaries on its articles,
- The Washington Post has a [tool](#) called “Ask the Post” that generates answers to questions from published Post articles

Questions: why deploy automated summaries ? If so, why LLM-based and not classical, extractive summaries, which are guaranteed from original source? What is the impact on long-term credibility?

#1 NEWS — AI is giving bad advice to flatter its users, says new study on dangers of overly agreeable chatbots

Link - <https://abcnews.com/Technology/wireStory/ai-giving-bad-advice-flatter-users-new-study-131443396>
<https://www.science.org/doi/10.1126/science.aec8352>

- **Context**
 - People are taking advice from LLM-based chatbots / AIs
 - LLM-based AI systems are being designed to be people pleasing
- **Findings**
 - Paper measured the prevalence of social sycophancy across 11 leading large language models. Compared results with what happens in similar situations on reddit
 - The model's responses were nearly 50% more sycophantic than humans', even when users engaged in unethical, illegal, or harmful behaviors.
 - **Participants' willingness to take responsibility and repair interpersonal conflicts decreases, while increasing their own conviction that they were right.**
- **Insight**
 - Sycophantic AI decreases prosocial intentions and promotes dependence
 - Users preferred and trusted sycophantic AI responses, incentivizing AI developers to preserve sycophancy despite the risks.

Update:

Administrative Information – CSCE 581

Walk through of
Github:

<https://github.com/biplav-s/course-tai-s26/>

Website:

<https://sites.google.com/site/biplavsrivastava/teaching/ai-csce-581-spring-2026-trusted-ai>

Quick Info - When and Where

- Tuesday/Thursday 11:40 am – 12:55 pm
- In person at **Swearingen Engr Ctr | Room 2A15**. Recordings will be available on Blackboard whenever feasible.

Catalog Information

- [Trusted AI - CSCE 581 001](#) CRN: 53067
- Duration: 01/12/2026 - 05/06/2026

Instructor Information

- Instructor: Biplav Srivastava, E-mail: [biplav.s AT sc.edu](mailto:biplav.s@sc.edu)
- Office Hours: 10:15-11:15am (Tu, Th); other times by appointment
- [GitHub](#) for slides, sample code.

TA Information

- Bharath Muppasani, bharath@email.sc.edu; Github ID – **BharathMuppasani**
- Office Hours: 11-12 (M), 2-3pm (W); Storey Building, Room 1209

Student Assessment

A = [920-1000]
B+ = [870-919]
B = [820-869]
C+ = [770-819]
C = [720-769]
D+ = [670-719]
D = [600-669]
F = [0-599]

Tests	Undergrad	Grad
Course Project – report, in-class presentation	600	600
Quiz – best of 2 from 3	200	200
Final Exam	200	100
Additional Final Exam – Paper summary, in-class presentation		100
Total	1000 points	1000 points

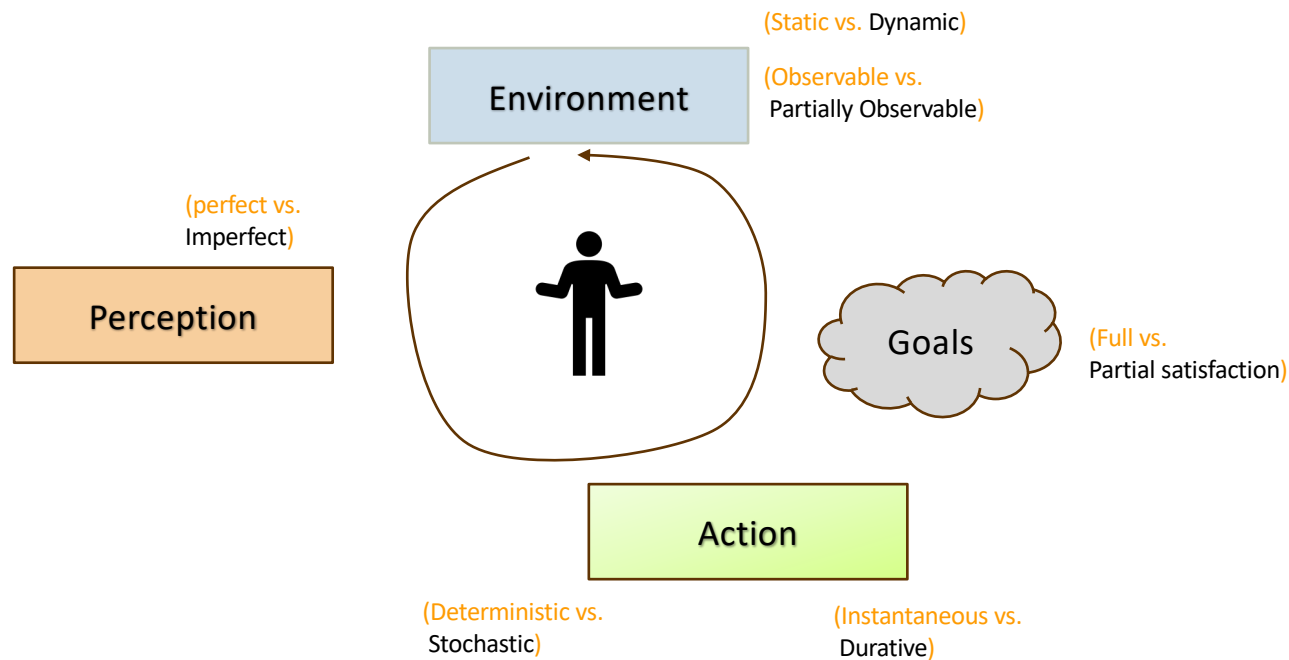
Grades and Attendance Policy

As an in-person class, I will impose a common-sense, **necessary – but not sufficient**, attendance policy for those taking the class for grades.

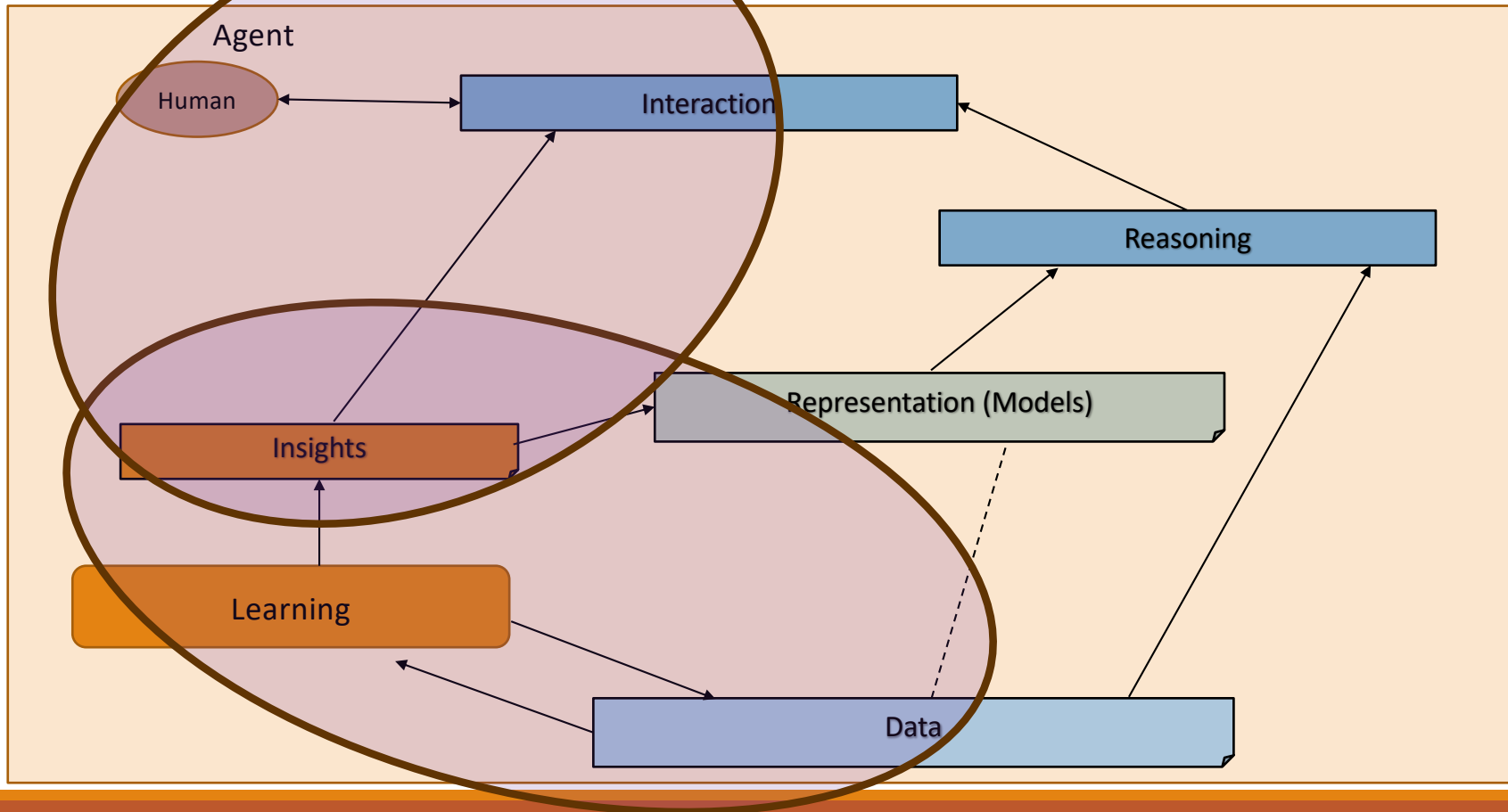
will be out of 14 classes:

1. 80% or more classes attending: **A, B+ or B**
2. 50-79% classes: **B+, B, C+, C, D+ or D**
3. 49% or below: **F**

Intelligent Agent Model



Relationship Between Main AI Topics (Covered in Course)



High Level Semester Plan (Adapted, Approximate)

CSCE 581 –

- Week 1: Introduction
- Week 2: Background: AI - Common Methods
- Week 3: The Trust Problem
- Week 4: Machine Learning (Structured data) - Classification
- Week 5: Machine Learning (Structured data) - Classification – Trust Issues
- Week 6: Machine Learning (Structured data) – Classification – Mitigation Methods
- Week 7: Machine Learning (Structured data) – Classification – Explanation Methods
- Week 8: Machine Learning (Text data, **vision**) – Classification,

Large Language Models

- Week 9: Machine Learning (Text data) - Classification – Trust Issues, LLMs
- Week 10: Machine Learning (Text data) – Classification – Mitigation Methods
- Week 11: Machine Learning (Text data) – Classification – Explanation Methods
- Week 12: Emerging Standards and Laws, **Real world applications**
- Week 13: Project presentations
- Week 14: Project presentations, Conclusion

AI/ ML topics and with a
focus on fairness, explanation,
Data privacy, reliability

Project Status Check and Timeline

- Project sprints
 - Second phase ending March 31 (Walk phase)
 - Third phase between April 2 to April 21 (Run phase)
- Project presentations
 - Apr 21 (Tu) Project presentation
 - Apr 23 (Th) Project presentation
- Project delivered
 - Apr 30 (Th) Project in Github (incl. report)

17	Mar 17 (Tu)	Invited Guest –	W10 Quiz 2 - start
18	Mar 19 (Th)	AI - Unstructured (Text): Processing and Representation	
19	Mar 24 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)	[Week 11] Quiz 2 - end
20	Mar 26 (Th)	Natural Languages/ Language Models and their Impact on AI	
21	Mar 31 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues	[Week 12] Project – Sprint 2 - end
22	Apr 2 (Th)	AI - Unstructured (Text): Analysis – Supervised ML – Mitigation Methods	Project – Sprint 3 - start
23	Apr 7 (Tu)	AI - Unstructured (Text): Analysis – Rating and Debiasing Methods	[Week 13] Quiz 3 - end
24	Apr 9 (Th)	Explanation Methods Trust: AI Testing	
25	Apr 14 (Tu)	Trust: Human-AI Collaboration	[Week 14] Quiz 3 - end
26	Apr 16 (Th)	Emerging Standards and Laws Trust: Data Privacy - Trusted AI for the Real World	
27	Apr 21 (Tu)	Project presentation	[Week 15] Project – Sprint 3 - end
28	Apr 23 (Th)	Project presentation	Last day of class
	Apr 28 (Tu)		Reading Day
29	April 30 (Th)	12:30pm – Final exam/ Overview	[Week 16] Optional, information shared

Introduction Section

Main Segment

Recap – ML, Classification / Supervised ML, Metrics

Machine Learning – Insights from Data

- Descriptive analysis
 - Describe a past phenomenon
 - **Methods:** classification (feedback from label), clustering, dimensionality reduction, anomaly detection, neural methods, reinforcement learning (feedback from hint/ reward)
- Predictive analysis
 - Predict about a new situation
 - **Methods:** time-series, neural networks
- Prescriptive analysis
 - What an agent should do
 - **Methods:** simulation, reinforcement learning, reasoning
- New areas
 - Counterfactual analysis
 - Causal Inferencing
 - Scenario planning

Concepts

- **Input data:** data available
 - **Training data:** used for training a learning algorithm and get a model
 - [Optional] **Validation data:** used to tune parameters
 - **Test data:** used to test a learning model
- **Classification problem**
 - Separating data into classes (also called labels, categorical types)
 - One of the attributes is the class label we are trying to learn
 - Class label is the **supervision**
- **Clustering problem**
 - We are trying to learn grouping of data
 - There is no attribute indicating membership in the groups (hence, **unsupervised**)
- **Prediction problem**
 - Learning value of a continuous variable

Reference: <https://machinelearningmastery.com/difference-test-validation-datasets/>
<https://www2.seas.gwu.edu/~bell/csci243/lectures/classification.pdf>

Many Method Types and Classifiers

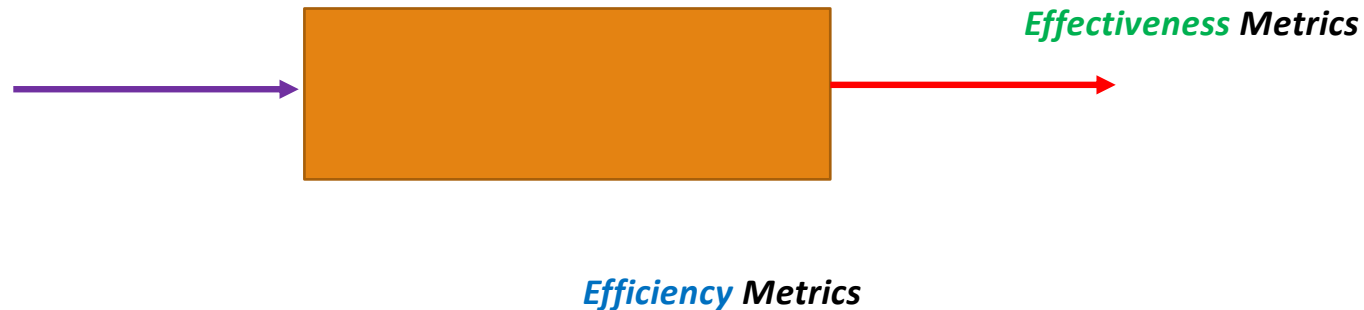
- Individual methods
 - Decision Tree (J48), R1, One-R
 - Naïve Bayes
 - ...
- Ensemble
 - Bagging: Aggregate classifiers (“bootstrap aggregation” => bagging)
 - Random Forest
 - Samples are chosen with replacement (bootstrapping), and combined (aggregated) by taking their average
 - Gradient Boosting: aggregate to turn weak learners into strong learners
 - Boosters (aggregators) turn weak learners into strong learners by focusing on where the individual weak models (decision trees, linear regressors) went wrong
 - Gradient Boosting

Source:

- Data Mining: Concepts and Techniques, by Jiawei Han and Micheline Kamber
- <https://towardsdatascience.com/getting-started-with-xgboost-in-scikit-learn-f69f5f470a97>

Metric Types

- **Effectiveness**: what the user of a system sees, primarily cares about
- **Efficiency**: what the executor in a system sees, primarily cares about



Metrics: Accuracy, Precision, Recall

Actual Class	Predicted class	
	Class = Yes	Class = No
Class = Yes	True Positive	False Negative
Class = No	False Positive	True Negative

Accuracy =
$$\frac{(TP+TN)}{(TP+FP+FN+TN)}$$

Precision =
$$\frac{(TP)}{(TP+FP)}$$

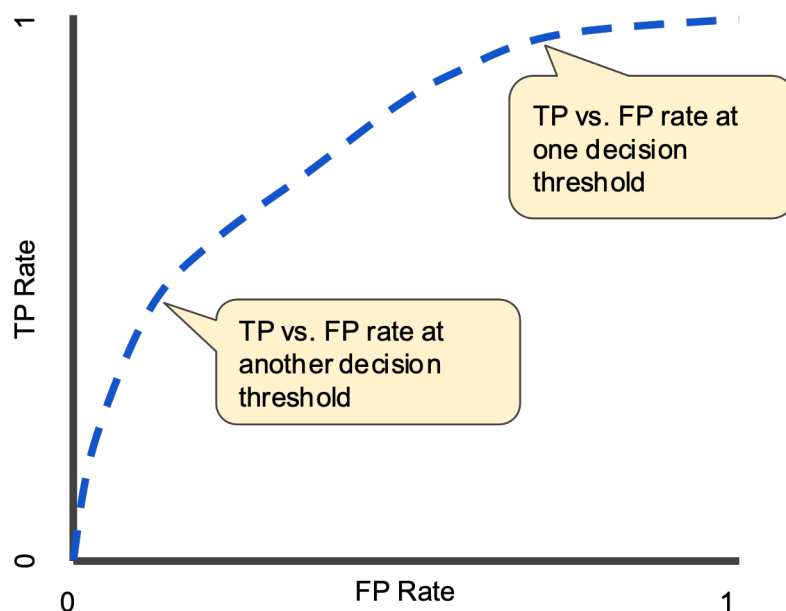
Recall =
$$\frac{(TP)}{(TP+FN)}$$

F1 Score: Harmonic Mean
$$\frac{1}{F1} = \frac{1}{Precision} + \frac{1}{Recall}$$

$$F1 = \frac{2 * (Recall * Precision)}{(Recall + Precision)}$$

ROC – Receiver Operating Characteristic curve

An ROC curve plots TPR vs. FPR at different classification thresholds



True Positive Rate = Recall =
$$\frac{(TP)}{(TP+FN)}$$

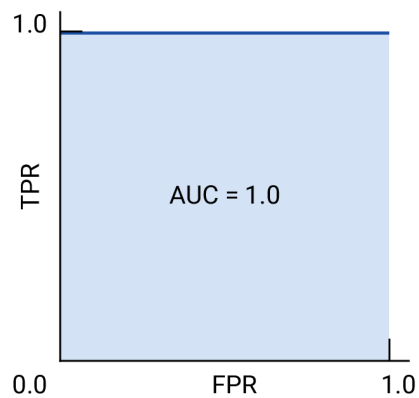
False Positive Rate =
$$\frac{(FP)}{(FP+TN)}$$

	Predicted class		
Actual Class		Class = Yes	Class = No
	Class = Yes	True Positive	False Negative
	Class = No	False Positive	True Negative

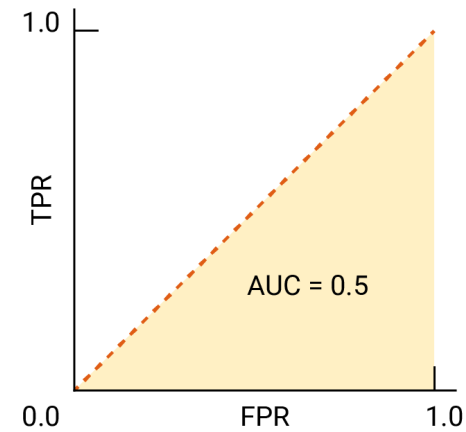
Source: <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>

AUC/ ROC Examples

ROC and AUC of a perfect system



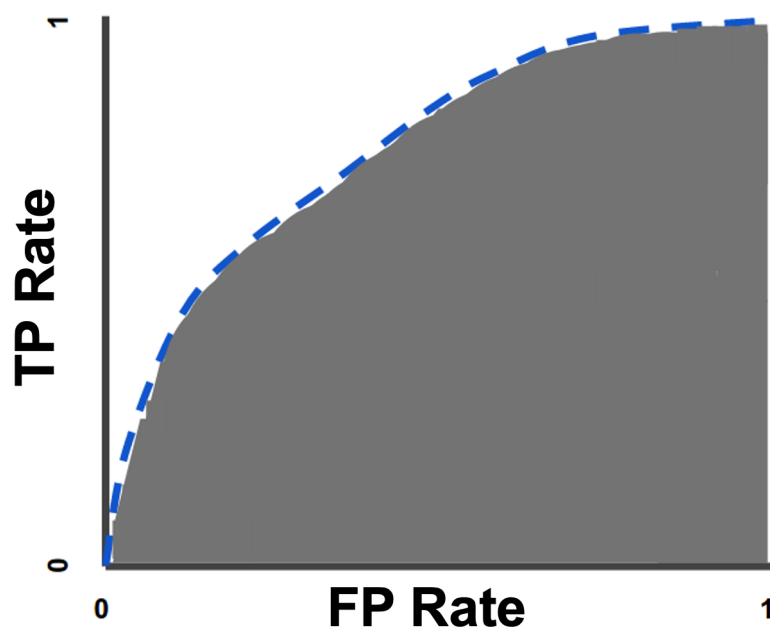
ROC and AUC of completely random guesses



The AUC is 0.5, representing a 50% probability of correctly ranking a random positive and negative example

Source: <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>

AUC – Area Under the ROC Curve



- Aggregate measure of performance across all possible classification thresholds.
- Interpretation: probability that the model ranks a random positive example more highly than a random negative example

Not helpful when the cost of false negatives vs. false positives are asymmetric

Source: <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>

Exercise and References

- Google:
<https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>
 - Take quiz
- Blogs: <https://blog.exsilio.com/all/accuracy-precision-recall-f1-score-interpretation-of-performance-measures/>

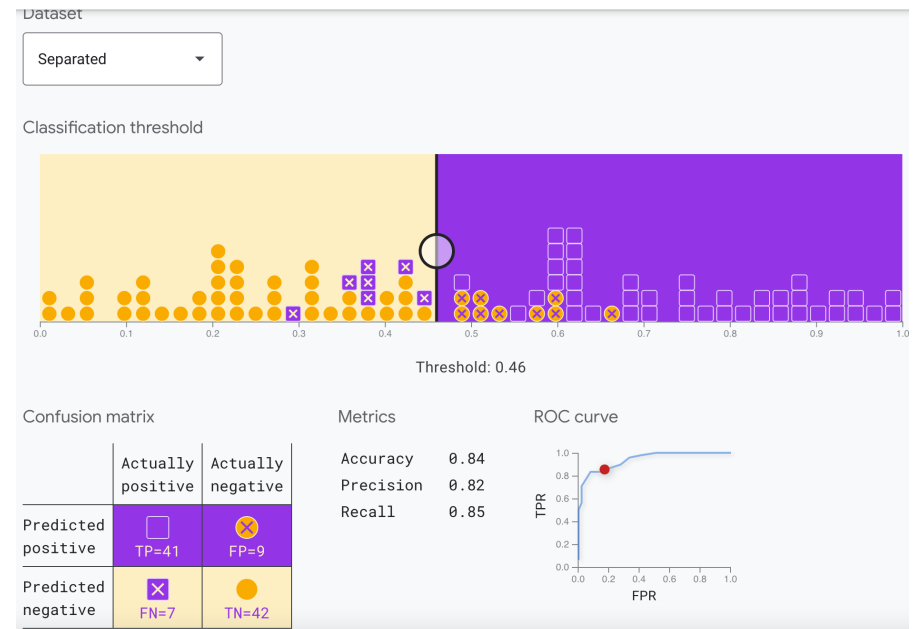
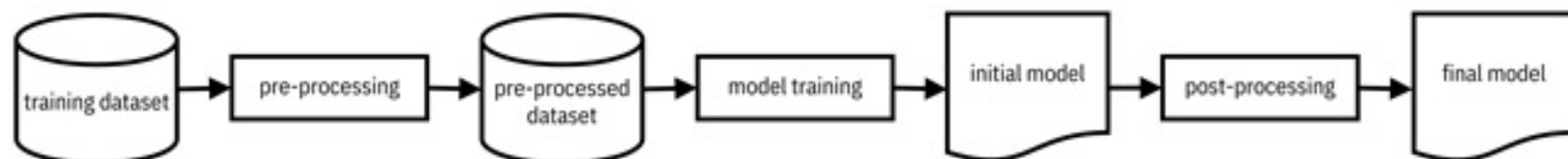


Image credit: <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>

So, What Changes With Text Classification?



Handling of (textual) data
before and after applying ML
methods!

Adapted from Image Credit: Trustworthy Machine Learning, Kush Varshney

Common NLP Tasks and Classification

- Extracting entities [Entity Extraction]
- Finding sentiment [Sentiment Analysis]
- Generating a summary [Text Summarization]
- Translating to a different language [Machine translation]
- Natural Language Interface to Databases [NLI]
- Natural Language Generation [NLG]

CSCE 771 goes into details

With resumes as data, what does this mean?

- Extracting entities [Entity Extraction]
e.g. colleges attended, work experience
 - Classification: is result (extraction) correct?
- Finding sentiment [Sentiment Analysis]
e.g. optimistic
 - Classification: assign sentiment label
- Generating a summary [Text Summarization]
e.g. 1-line summary
 - Classification: is result (summary) correct with respect to length?
- Translating to a different language [Machine translation]
e.g. Spanish
 - Classification: is result (translation) correct?
- Natural Language Interface to Databases [NLI]
e.g. candidate search
- Natural Language Generation [NLG]
e.g. cover letter generation

Which ML Classification Method to Choose?

- Blog: “Crawl – Walk – Run”
 - [The Crawl-Walk-Run Approach for AI-based Real World Problem Solving](#), Feb 2025, Biplav Srivastava
- Reading material:
 - “Which ML to Use” with title: Data-driven advice for applying machine learning to bioinformatics problems
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5890912/>
 - “10 tips with title”: Ten quick tips for machine learning in computational biology
<https://biodatamining.biomedcentral.com/articles/10.1186/s13040-017-0155-3>

Discussion: 10 Tips Paper

- Access: <https://biodatamining.biomedcentral.com/articles/10.1186/s13040-017-0155-3>
- Chicco, D. Ten quick tips for machine learning in computational biology. *BioData Mining* **10**, 35 (2017). <https://doi.org/10.1186/s13040-017-0155-3>

The Tips

- Tip 1: Check and arrange your input dataset properly
- Tip 2: Split your input dataset into three independent subsets (training set, validation set, test set), and use the test set only once you complete training and optimization phases
- Tip 3: Frame your biological problem into the right algorithm category
- Tip 4: Which algorithm should you choose to start? The simplest one!
- Tip 5: Take care of the imbalanced data problem
- Tip 6: Optimize each hyper-parameter
- Tip 7: Minimize overfitting
- Tip 8: Evaluate your algorithm performance with the Matthews correlation coefficient (MCC) or the Precision-Recall curve
- Tip 9: Program your software with open source code and platforms
- Tip 10: Ask for feedback and help to computer science experts, or to collaborative Q&A online communities

Examples of Classification with Text

- Sentiment analysis (*assign sentiment classes*)
- Annotation (*assigning entity type to text*)
- Application specific
 - Fake news
 - Spam email
 - ...

When to Use Learning/ When Not

- When
 - There are too many patterns to enumerate
 - When data is plentifully available
 - [**Cost**: annotation to know the class labels, re-learning to avoid model shift]
- When Not
 - There are few, distinct patterns that can be easily enumerated
 - When there may be a high cost of accessing data
 - When there is a need to exert control on the AI development process; e.g., sensitive domain
 - [**Cost**: of knowing the patterns completely]

ML – Supervised (Fake News)

- By Example:
 - <https://github.com/biplav-s/course-nl/blob/master/I9-ml-review/Classification%20-%20Fake%20news.ipynb>
- Fake news dataset

In-Class Tasks

- On your resume, do these tasks
 - [Visual summary] Create a word tag cloud of text
 - [Entity extraction] Identify universities attended
 - [Sentiment] Find the sentiment (emotion) that may be ascribed
 - [1-line Summary] Generate a 1-line summary with a LLM
- On the set of resumes from class,
 - [Visual summary] Create a word tag cloud of the corpus
 - [Entity extraction] Identify universities attended by all students
 - [Sentiment] Find the sentiment (emotion) that may be ascribed to the corpus
 - [1-line Summary] Generate a 1-line summary with the corpus
- Compare yourself with the class
 - Where are the differences, if any?

Reference notebooks/ hints:

1. **Word-tag** - https://github.com/biplav-s/course-nl-f22/blob/main/sample-code/l1-wordcloud/FirstLook_ClassSyllabusData.ipynb
2. **Entity: use spacy** - <https://github.com/biplav-s/course-nl-f22/blob/main/sample-code/l17-eventextr/SimpleEntitySearch.ipynb>
3. **Sentiment: use lexicon method** - <https://github.com/biplav-s/course-d2d-ai/blob/7f90f154729115a31f449702dbdf84d63be7a844/sample-code/l23-textrepresent/Basic%20Sentiment.ipynb>
4. **Optional: Use BERT/ distilBERT**

Take Home Task

- Apart from Logistic Regression and Random Forest, create another model
- Run results for title and total text
- Report accuracy, F1 and AUC ROC
- Recommend a model to use for the task (fake news classification) based on your results

ML – Supervised (Movie Sentiment)

- By Example:
 - Data: IMDB – 50K movie reviews
<http://ai.stanford.edu/~amaas/data/sentiment/>
<https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews/>
- Sample code
 - Classical ML methods:
 - <https://www.kaggle.com/code/eminaanapaydn/imdb-sentiment-analysis> (MultinomialNB)
 - <https://www.kaggle.com/code/youssefemad004/sentimentanalysisml> (MultinomialNB, RandomForest)

Where Can LLMs be Used in Classification (Text)

- Text (word) processing
- Classification task // learn to assign classes
- Evaluation ...

ML – Supervised (Movie Sentiment)

- By Example:

- Data: IMDB – 50K movie reviews

- <http://ai.stanford.edu/~amaas/data/sentiment/>

- <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews/>

- Sample code

- Classical ML methods:

- <https://www.kaggle.com/code/eminaanapaydn/imdb-sentiment-analysis> (MultinomialNB)

- <https://www.kaggle.com/code/youssefemad004/sentimentanalysisml> (MultinomialNB, RandomForest)

- LLM/ BERT classifier:

- <https://www.kaggle.com/code/tcc3281/bert-sentiment>

ML – Supervised (Movie Genre)

- Data: IMDB – 5K movie reviews, 10 columns
<https://www.kaggle.com/datasets/rakkesharv/imdb-5000-movies-multiple-genres-dataset>
- Task: Assigning genre
- Sample Code: <https://www.kaggle.com/code/yazeedalsahouri03/imdb-movies-prediction>

ML – Supervised (Mental Health)

- Example 1:
 - Data - Chat: <https://www.kaggle.com/datasets/elvis23/mental-health-conversational-data>
 - Classification Task: Classify intent, and then return corresponding response
 - Code - <https://www.kaggle.com/code/selimkhaled50/complete-mental-health-chatbot>
- Example 2:
 - Data: <https://www.kaggle.com/datasets/bhavikjikadara/mental-health-dataset> (300k rows and 17 columns)
 - Classification Task: Social_weakness
 - Code - <https://www.kaggle.com/code/mahmoudweso/mental-health-dataset>

Real World Application

Amazon Ads is committed to protect customers from fraud, abuse, illegal and harmful content. We prioritize customer protection through a comprehensive approach of strict policies, thorough review processes, and safety measures to ensure digital advertising content meet high standards. We maintain robust reporting systems, allowing for swift action against policy violations. These multilayered safeguards work in tandem to shield customers from exposure to abusive and harmful content, fostering a secure advertising environment for both consumers and advertisers.

We welcome proposals in the context of digital advertising in the following research tracks:

1. Fraud, Abuse, and financial scams
2. Behavioral foundational models to distinguish human and bot behavior
3. Multi-modal classification of online content and websites
4. Device and website identification
5. Detection of plagiarized content
6. Cyber threat activity, malicious and adversarial misuse of AI and LLMs
7. Decentralized identifiers, verifiable credentials, data lineage and transparency
8. Large language models for labeling, annotation and auditing of model performance
9. Detection and mitigation of hallucination in LLMs for content analysis
10. Protection of vulnerable individuals, specially child and teen safety

Funder: Amazon

Website: <https://www.amazon.science/research-awards/call-for-proposals/amazon-ads-call-for-proposals-spring-2025>

Trust Issue – Stability of Output

Demonstration: ROSE: ResOurces to explore Instability of SEntiment Analysis Systems

ROSE: tool and data ResOurces to explore the instability of SEntiment analysis systems

Explore emotions by words (positive, negative)

Explore emotions by pronouns (one by one)

Explore emotions by pronouns (all at once)

Explore emotions by proper nouns (one by one)

Explore emotions by proper nouns (all at once)



Scan the code to
try our ROSE
tool!

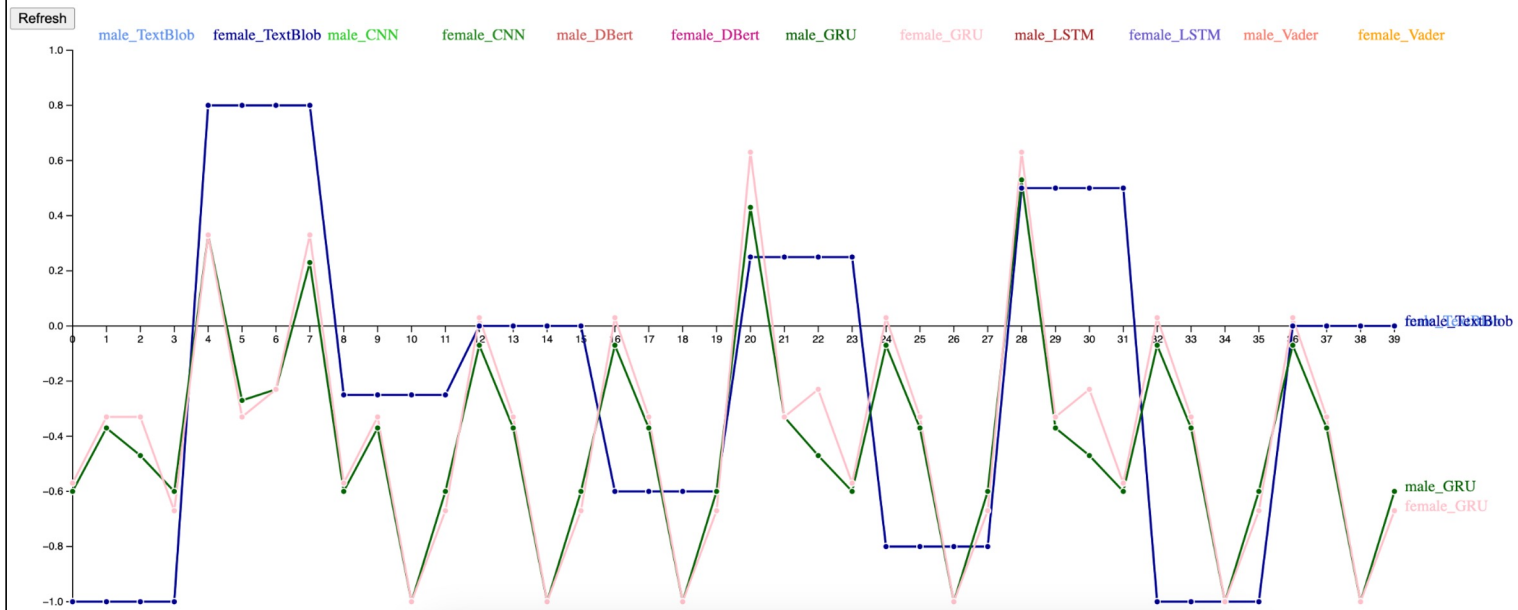
References:

1. MUNDADA, GAURAV, KAUSIK LAKKARAJU, and BIPLAV SRIVASTAVA. "ROSE: Tool and Data ResOurces to Explore the Instability of SEntiment Analysis Systems."

Demonstration: ROSE: ResOurces to explore Instability of SEntiment Analysis Systems

Average Sentiment Scores for Proper Nouns (all at once)

- Click on any SAS below to see the visualization of sentiment scores for that SAS
- Click on the 'Refresh' button below to remove all the graphs
- Hovering over a data point shows the sentence it denotes (at the bottom of the page)
- Y-axis denotes the sentiment score of that sentence



References:

1. MUNDADA, GAURAV, KAUSIK LAKKARAJU, and BIPLAV SRIVASTAVA. "ROSE: Tool and Data ResOurces to Explore the Instability of SEntiment Analysis Systems."

Instability of AI is Well Recorded

- [Text] [Su Lin Blodgett](#), [Solon Barocas](#), [Hal Daumé III](#), [Hanna Wallach](#), Language (Technology) is Power: A Critical Survey of “Bias” in NLP, Arxiv - <https://arxiv.org/abs/2005.14050>, 2020 [NLP Bias]
- [Image] Vegard Antun, Francesco Renna, Clarice Poon, Ben Adcock, and Anders C. Hansen, On instabilities of deep learning in image reconstruction and the potential costs of AI, <https://doi.org/10.1073/pnas.1907377117>, PNAS, 2020
- [Audio] Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R. Rickford, Dan Jurafsky, and [Sharad Goel](#), Racial disparities in automated speech recognition, PNAS April 7, 2020 117 (14) 7684-7689, <https://doi.org/10.1073/pnas.1915768117>, March 23, 2020

Project Discussion

Project Milestones and Deliverables

- See course website for schedule and milestones:
<https://sites.google.com/site/biplavsrivastava/teaching/ai-csce-581-spring-2026-trusted-ai>
 - Sprints: 1 (Crawl) : Feb 5-24, 2(Walk) : Feb 24-Mar31 and 3 (Run): Apr 2-21
- Milestones – in-class presentations
 - Project discussion 1 – Feb 5
 - Project discussion 2 – March 5
 - **Project presentations 3 (final) – April 21 and 23**
 - **Project complete by April 30, 2026 in Github**
 - **Report, code, presentation (for completeness)**

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
- Timelines of reports [-10 points]
- Code missing [-10 points]
- Testcase missing [-10 points]

Rubric for Evaluation of Course Project

Project

- Project plan along framework introduced (7 points)
- Challenging nature of project
- Actual achievement
- Report
- Sharing of code

Presentation

- Motivation
- Coverage of related work
- Results and significance
- Handling of questions

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
- Timelines of reports [-10 points]
- Code missing [-10 points]
- Testcase missing [-10 points]

Course Project

- **Framework**

1. (Problem) Think of a problem whose solution may benefit people (e.g., health, water, air, traffic, safety)
2. (User) Consider how the primary user (e.g., patient, traveler) may be solving the problem today
3. (AI Method) Think of what the solution will do to help the primary user
 1. Solution => ML task (e.g. classification), recommendation, text summarization, ...
 2. Use a foundation model (e.g., LLM-based) solution as the baseline
4. (Data) Explore the data for a solution to work
5. (Reliability: Testing) Think of the evaluation metric we should employ to establish that the solution will work? (e.g., 20% reduction in patient deaths)
6. (Holding Human Values) Discuss if there are fairness/bias, privacy issues?
7. (Human-AI) Finally, elaborate how you will explain the primary user that your solution is trustable to be used by them

Project Discussion: What to Focus on ?

- Problem: you should care about it
- Data: should be available
- Method: you need to be comfortable with it. Have at least two – one serves as baseline
- Trust issue
 - Due to Users
 - Diverse demographics
 - Diverse abilities
 - Multiple human languages
 - Or other impacts
- What one does to mitigate trust issue

Project Discussion

1. Create a private Github repository called “CSCE581-Spring2025-<studentname>-Repo”. Share with Instructor (biplav-s)
2. Create a folder called “Project”. Inside, create a text file called “ProjectPlan.md” (or “ProjectPlan.txt”) and have details by the next class (Jan 30, 2026)

1. Title:
2. Key idea: (2-3 lines)
3. Who will care when done:
4. Data need:
5. Methods:
6. Evaluation:
7. Users:
8. Trust issue:

Concluding Section

Week 11 (L21 and 22): Concluding Comments

- We looked at
 - L21: Supervised ML (Text)
 - L22: Classification, Trust Issue

About Next Week – Lectures 23, 24

Lectures 23, 24

- Trust issue – Mitigation – AI Explanation (XAI)
- Trust Issues – Mitigation – AI Rating/ Certification

17	Mar 17 (Tu)	Invited Guest –	W10 Quiz 2 - start
18	Mar 19 (Th)	AI - Unstructured (Text): Processing and Representation	
19	Mar 24 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)	[Week 11] Quiz 2 - end
20	Mar 26 (Th)	Natural Languages/ Language Models and their Impact on AI	
21	Mar 31 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues	[Week 12] Project – Sprint 2 - end
22	Apr 2 (Th)	AI - Unstructured (Text): Analysis – Supervised ML – Mitigation Methods	Project – Sprint 3 - start
23	Apr 7 (Tu)	AI - Unstructured (Text): Analysis – Rating and Debiasing Methods	[Week 13] Quiz 3 - end
24	Apr 9 (Th)	Explanation Methods Trust: AI Testing	
25	Apr 14 (Tu)	Trust: Human-AI Collaboration	[Week 14] Quiz 3 - end
26	Apr 16 (Th)	Emerging Standards and Laws Trust: Data Privacy - Trusted AI for the Real World	
27	Apr 21 (Tu)	Project presentation	[Week 15] Project – Sprint 3 - end
28	Apr 23 (Th)	Project presentation	Last day of class
	Apr 28 (Tu)		Reading Day
29	April 30 (Th)	12:30pm – Final exam/ Overview	[Week 16] Optional, information shared